Lab 03 Instructions

Due Feb 10 5:00 PM PST

STAT240 Spring 2026

To solve these problems please submit a pdf generated from a qmd or Rmd file, showing all code required to solve the problems. You can use the template `Lab_03.qmd`. For any text answers please type them directly (making them bold), do not use `cat` inside an R code chunk, as these do not render clearly.

Note that if you run the required `sql` commands on the database multiple times you will likely repeatedly update the underlying data. This is not a problem for this (simple) example, and you should get the same solutions when you look at the last few rows regardless of this. Don't worry if the number of rows increases for example.

Problem 1: Hazardous waste

The material for this lab includes data about hazardous waste in Austria, Armenia and Azerbaijan acquired from data.un.org. In the sqlite file combined.sqlite a table called 'lab03combined' is provided. This table has 6 columns: 'Country', 'Year', 'Generated', 'Incinerated', 'Landfilled', 'Recycled'. This table indicates the amount of hazardous waste generated, incinerated, landfilled or recycled (in tonnes) in the corresponding year and country, according to the UNSD/UNEP Questionnaire on Environment Statistics. For example, the first row of this table is:

Country	Year	Generated	Incinerated	Landfilled	Recycled
Armenia	1998	286,607	0	286,607	0

This row indicates that in 1998 in Armenia, 286,607 tonnes of hazardous waste were generated according to the survey. The years provided in the sqlite file span 1995 to 2012. Note that not all countries have the same set of provided years.

Part a)

As a first initial task you should take the data from `incinerated.new`, a csv file, and create a new table from this data. Write the first **4** rows of this dataset to a new table in the database, calling this table `incin`.

(2 points)

```
## write answer here
```

Part b)

Now use `dbExecute` and `INSERT` to add the final three rows of `incinerated.new` to this table `incin`. Print out the final three rows of `incin` to confirm you have done this correctly.

(2 points)

```
## write answer here
```

Part c)

Your next task is to update the original table in the database with all the new, more recent data. The new data are provided in the comma separated variable files `generated.new, incinerated.new, landfilled.new, and recycled.new` (corresponding to the respective columns of the sqlite table). Write a R script that will update the sqlite table with these new data by inserting the new data to the end of the table. One new row must be inserted into the sqlite table for each unique Country/Year pair.

You can assume (initially) that each of these files contains the same number of rows, and that each file will contain the same value of country and year in the same rows.

For example, since the first row of data in `generated.new` is:

```
Armenia,2015,555075
```

and the first row of `incinerated.new` is:

```
Armenia,2015,3767
```

a new row must be inserted into the combined.sqlite `lab03combined` table with Country value `Armenia`, Year value `2015`, Generated value `555,075` and Incinerated value `3,767` (and so on for Landfilled and Recycled, drawn from their corresponding .new files).

Write your code using `dbExecute` and `INSERT` statements. Do not hard code the `INSERT` statements (i.e., if new `.new` files were provided again 5 years from now with the same format and filenames, rerunning your code should further update the database to insert the next version of the data, regardless of how many rows are in the new `.new` files.)

(4 points)

```
## write answer here
```

Part d)

Print the last 7 rows of the updated sqlite table. You can do this by dumping the updated sqlite table into a R data frame, and then printing the last 7 rows of the data frame.

(2 points)

```
## write answer here
```

Part e) (Optional Bonus)

We have made the assumption here that the data is in a nice format with the same ordering in all the files. In reality you will never be this lucky. Briefly think about how you would complete part c if there was not a consistent ordering in all the csv files. Code is not required, you can just give some ideas!